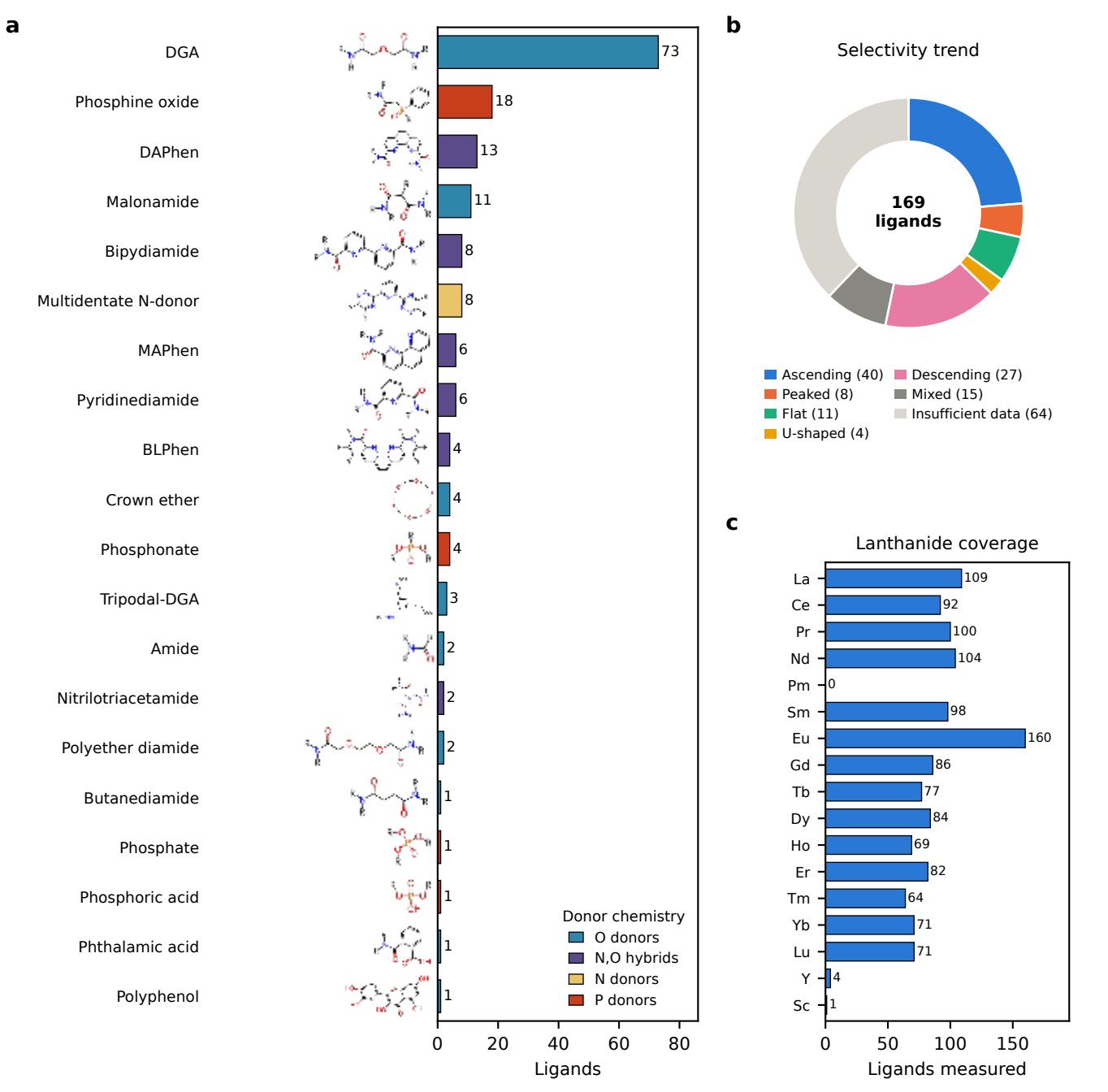
a
DGA
Phosphine oxide
DAPhen
Malonamide
Bipydiamide
Multidentate N-donor
MAPhen
Pyridinediamide
BLPhen
Crown ether
Phosphonate
Tripodal-DGA
Amide
Nitrilotriacetamide
Polyether diamide
Butanediamide
Phosphate
Phosphoric acid
Phthalamic acid
Polyphenol
73
18
13
11
8
8
6
6
4
4
4
3
2
2
2
1
1
1
1
1
Donor chemistry
O donors
N,O hybrids
N donors
P donors
0
20
40
60
80
Ligands
b
Selectivity trend
169 ligands
Ascending (40)
Descending (27)
Peaked (8)
Mixed (15)
Flat (11)
Insufficient data (64)
U-shaped (4)
c
Lanthanide coverage
La 109
Ce 92
Pr 100
Nd 104
Pm 0
Sm 98
Eu 160
Gd 86
Tb 77
Dy 84
Ho 69
Er 82
Tm 64
Yb 71
Lu 71
Y 4
Sc 1
0
50
100
150
Ligands measured